CONNECT'VILLE - ARCHITECTURE BASE DE DONNÉES COMPLETE

1. PRINCIPES GENERAUX

```
Architecture multi-tenant basée sur PostgreSQL.
Chaque enregistrement contient city_id.
Isolation logique par ville.
Indexation avancée pour performance.
RGPD compliant avec anonymisation possible.
```

2. TABLE USERS

```
CREATE TABLE users (
    id UUID PRIMARY KEY,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash TEXT,
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    phone VARCHAR(30),
    city_id UUID NOT NULL,
    role VARCHAR(20) DEFAULT 'citizen',
    verified BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

3. TABLE CITIES

```
CREATE TABLE cities (
    id UUID PRIMARY KEY,
    name VARCHAR(255),
    postal_code VARCHAR(10),
    country VARCHAR(100),
    branding JSONB,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

4. TABLE MERCHANTS

```
CREATE TABLE merchants (
    id UUID PRIMARY KEY,
    city_id UUID REFERENCES cities(id),
    name VARCHAR(255),
    description TEXT,
    address TEXT,
    category VARCHAR(100),
    subscription_level VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. TABLE CIVIC_WALLETS

```
CREATE TABLE civic_wallets (
    user_id UUID PRIMARY KEY REFERENCES users(id),
    available_balance INT DEFAULT 0,
    total_earned INT DEFAULT 0,
    total_spent INT DEFAULT 0,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

6. TABLE CIVIC_TRANSACTIONS

```
CREATE TABLE civic_transactions (
    id UUID PRIMARY KEY,
    user_id UUID REFERENCES users(id),
    city_id UUID REFERENCES cities(id),
    type VARCHAR(50),
    amount INT,
    mission_id UUID,
    merchant_id UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

7. TABLE MISSIONS

```
CREATE TABLE missions (
    id UUID PRIMARY KEY,
    city_id UUID REFERENCES cities(id),
    title VARCHAR(255),
    description TEXT,
    reward_amount INT,
    validation_type VARCHAR(50),
    active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

8. TABLE REWARDS

```
CREATE TABLE rewards (
    id UUID PRIMARY KEY,
    merchant_id UUID REFERENCES merchants(id),
    city_id UUID REFERENCES cities(id),
    title VARCHAR(255),
    cost INT,
    max_claims INT,
    expiration_date TIMESTAMP
);
```

9. INDEX & PERFORMANCE

```
CREATE INDEX idx_users_city ON users(city_id);
CREATE INDEX idx_transactions_user ON civic_transactions(user_id);
CREATE INDEX idx_missions_city ON missions(city_id);
Partition possible par region pour scalabilité Europe.
```